

The empirical recurrence for OEIS A259001, proved by transfer matrix

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Abstract

OEIS A259001 counts the arrays over $\{0, \dots, 1\}$ whose 3×3 subblocks carry a statistic that is monotone along named directions of the subblock grid, and carries an empirical recurrence of order 22 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Every direction named moves the subblock row by at most one, so the count is a walk count on $S = 6788$ states — the last two array rows together with the subblock row just completed — and the conjectured recurrence is then settled by a finite exact computation.

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1 The conjecture

OEIS A259001 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock sum of the two medians of the central row and column plus the two sums of the diagonal and antidiagonal nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

9374, 18341, 55760, 116098, 142316, 191035, 134036, 138255, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = -a(n-1) - a(n-2) + a(n-3) + a(n-4) + 2*a(n-5) + a(n-6) + a(n-7) - a(n-8) - a(n-9) - a(n-10) + a(n-12) + a(n-13) + a(n-14) - a(n-15) - a(n-16) - 2*a(n-17) - a(n-18) - a(n-19) + a(n-20) + a(n-21) + a(n-22)$ for $n > 36$.

As of the “Last modified” line on the live entry (Nov 16 2015 04:45:03, revision 12) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

A 3×3 subblock has four distinguished lines of three cells: its diagonal, its antidiagonal, its central row and its central column. The entry’s statistic is a signed sum of aggregates — sums, maxima, minima, medians — taken over those lines, and $\sigma(i, j)$ denotes its value on the subblock in position (i, j) of the subblock grid, which has n rows and 3 columns for an $(n+2) \times 5$ array over $\{0, \dots, 1\}$. The condition is

$$\sigma(i, j) \leq \sigma(i, j+1), \quad \sigma(i, j) \leq \sigma(i+1, j), \quad \sigma(i, j) \leq \sigma(i+1, j-1)$$

whenever both positions exist.

“ne-to-sw antidiagonally” is the offset $(+1, -1)$ on the subblock grid; that reading, and the reading of every statistic name here, was checked by counting the small arrays directly against what the entries publish.

3 The count is a walk count

A subblock spans three array rows and the vertical comparison spans four, so the obvious state is three consecutive rows. It is larger than it needs to be. What a step actually consults is the last TWO array rows — enough to build the next subblock row once one more row arrives — and the PREVIOUS SUBBLOCK ROW, to compare against. So take as state the triple

$$(r_{i+1}, r_{i+2}, \sigma\text{-row } i),$$

and let a step append an array row r_{i+3} : that fixes σ -row $i + 1$ from $r_{i+1}, r_{i+2}, r_{i+3}$, and every comparison between the two σ -rows is settled there. All the third rows that produce the same σ -row collapse into one state, which is what makes the wider arrays reachable.

The starting vector is therefore weighted rather than all-ones: a state is entered by as many first rows as produce it, and ι records that count. With τ all-ones, an $(n + 2)$ -row array is a walk of $n - 1$ steps and

$$a(n) = \iota^\top M^{n-1} \tau$$

on the $S = 6788$ reachable states. In particular a is C -finite. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^{22} - \sum_i c_i t^{22-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+22} - \sum_i c_i M^{j+22-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = -a(n-1) - a(n-2) + a(n-3) + a(n-4) + 2a(n-5) + a(n-6) + a(n-7) - a(n-8) - a(n-9) - a(n-10) + a(n-11)$$

for every $n > 36$.

Proof. The vector $w = q(M) \tau$ was formed by 22 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 36 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 30 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.